KAMIL MAJERI - END-TO-END POWER BI - DATA PIPELINE

Two years of a UK online gift shop, from raw data to decisions

Over one million invoice lines from the Online Retail II dataset, cleaned in Python, stored in PostgreSQL and analysed in Power BI. Use the tiles below to explore revenue, products, customers, markets and data quality.

£18,9M
Net Revenue

39 515
Orders

5 853
Known customers
≈ 6,321 in total

4 722
Products Sold

How the data flows

01 Extract
PYTHON & KAGGLEHUB

Downloads the Online Retail II CSV from Kaggle: over one million invoice lines.

→

02 Transform
PYTHON & PANDAS

13 cleaning steps: types, duplicates, non-products, error lines, outliers, etc.

→

03 Load
POSTGRESQL & DOCKER

Star schema: a sales fact table linked to product and customer dimension tables.

→

04 Analyse
POWER BI & DAX

Data model with DAX measures, KPI cards and five analysis pages.

Explore the report

Overview
How is the business doing?
Open →

Products
What drives revenue?
Open →

Customers
Do customers come back?
Open →

Countries & Time
Where and when do we sell?
Open →

Data Quality
Can we trust the numbers?
Open →

How is the business doing?

Amounts in GBP. Data runs from 1 Dec 2009 to 9 Dec 2011. 7% of orders have no customer ID.

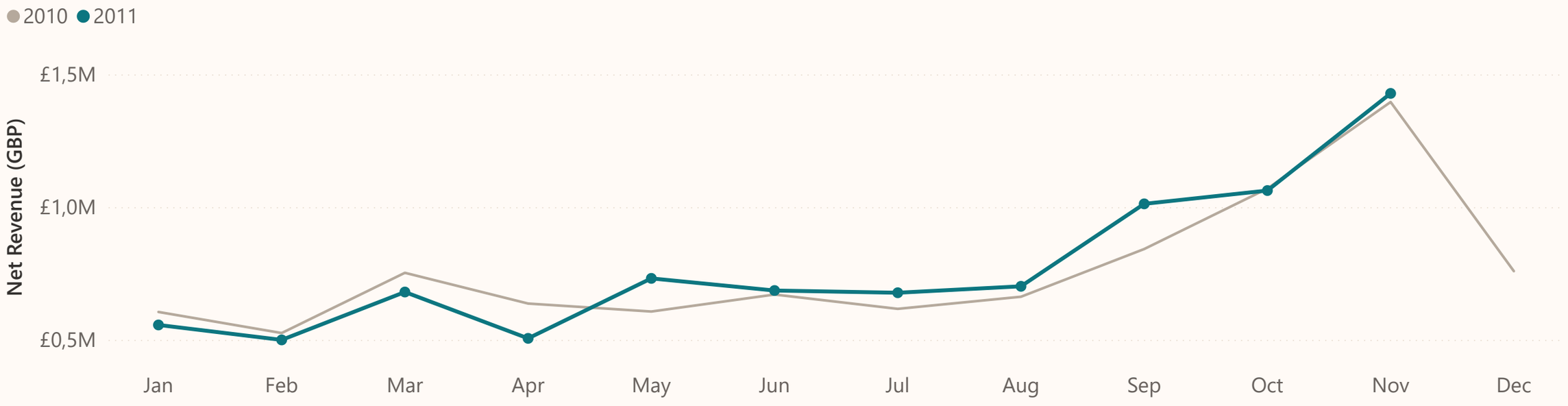
Year: All

Country: All

£18,9M	39 515	5 853	£489	2,3%	1,9%
Net Revenue	Orders	Known customers	Avg Order Value	Refund Rate	Growth vs 2010

Sales peak every November, ahead of Christmas

Net Revenue by month. December 2011 is not shown: data stops on 9 December.



What drives revenue?

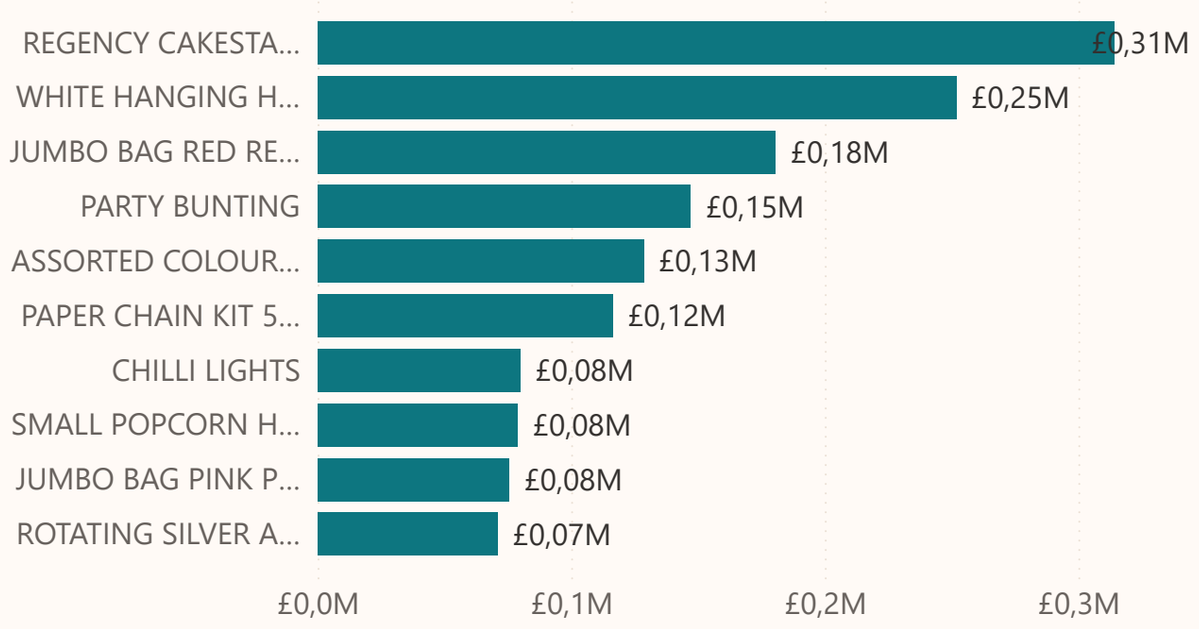
This looks like a wholesale business: cheap gift items bought in bulk by shops. A small part of the catalogue makes most of the revenue; the rest sells little.

Year: All

Country: All

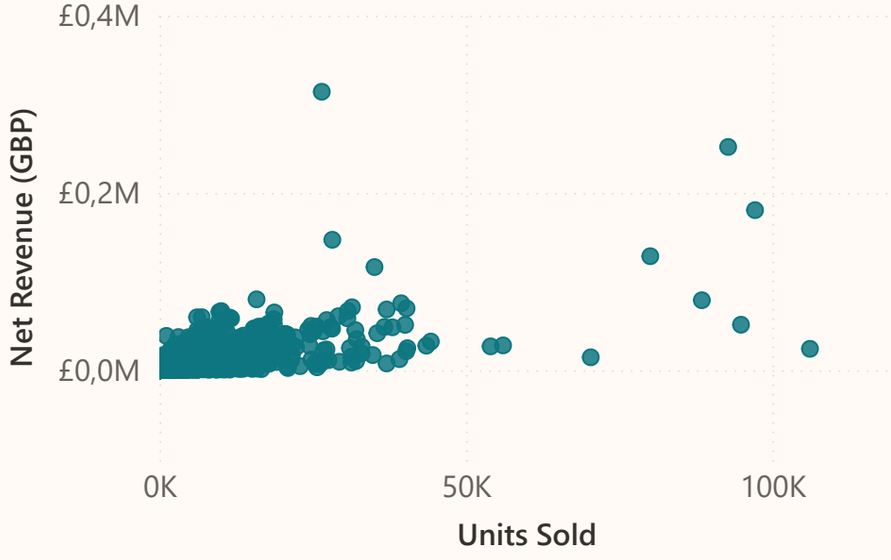
4 722	11 015 095	£1,76	279
Products Sold	Units Sold	Avg Unit Price	Units per Order

The best-seller makes less than 2% of Net Revenue



A few products sell far more than the rest

Higher = more revenue. Further right = higher sales volume. Most products are concentrated in the bottom-left.



Kamil Majeri / Source: Online Retail II (via Kaggle)

Do customers come back?

Retention = share of a quarter's customers who buy again in the next quarter. Identified customers only, whole period, all countries.

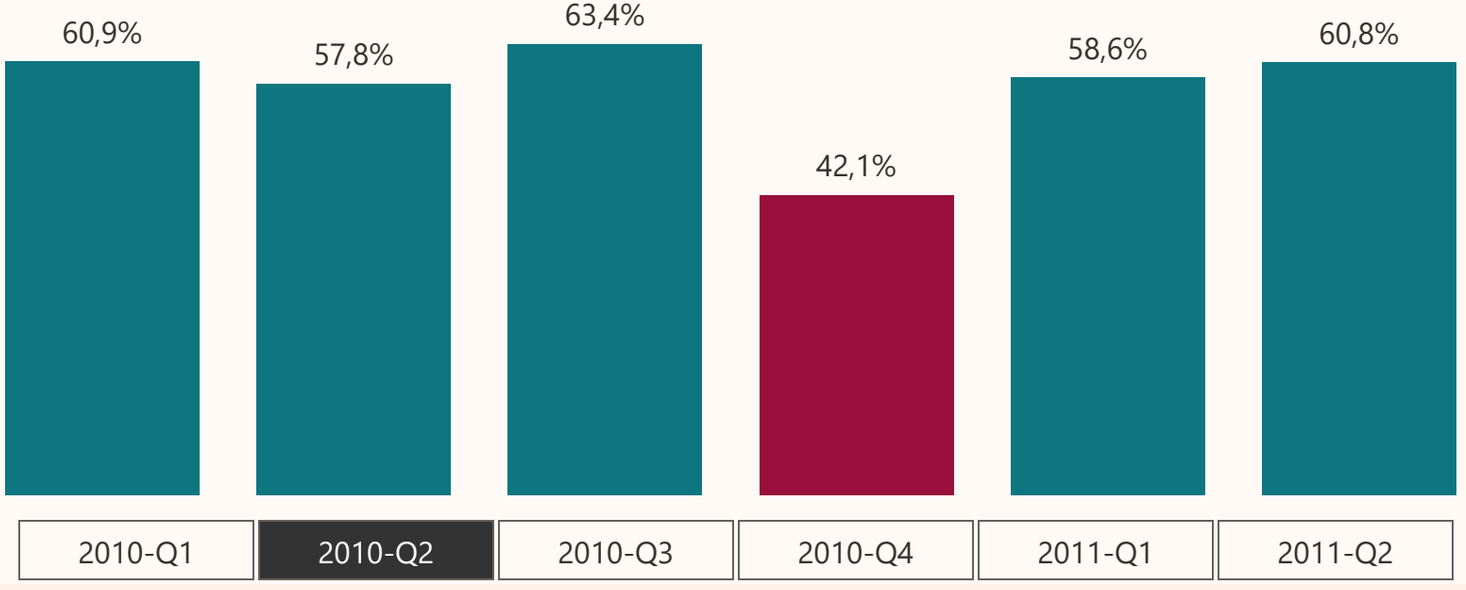
2 040
Customers in quarter

1 179
Customers back next quarter

57,8%
Quarterly Retention

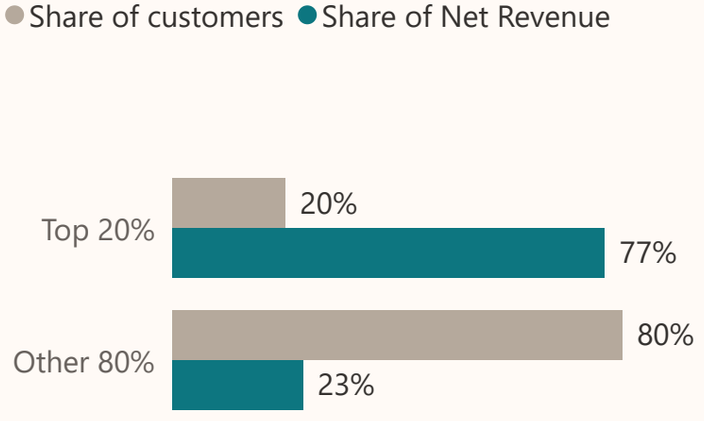
About 6 in 10 customers buy again the next quarter

Christmas buyers rarely come back next quarter. Click a quarter below to update the cards.



The top 20% of customers bring in about three quarters of Net Revenue

The top 1% of customers alone brings in 31%.



Kamil Majeri / Source: Online Retail II (via Kaggle)

Where and when do we sell?

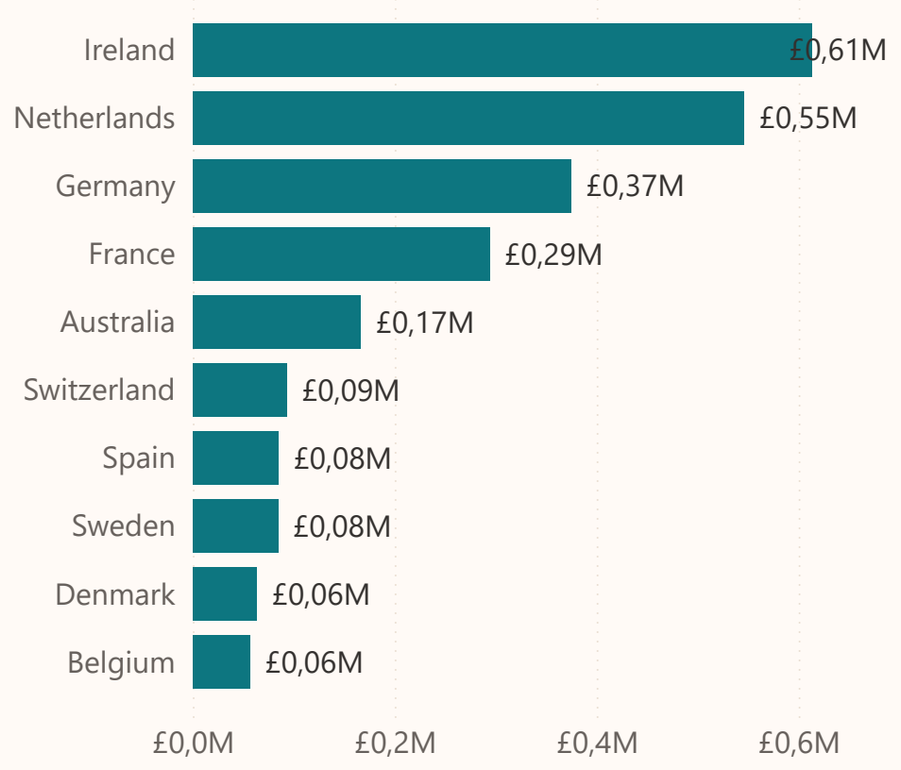
The UK makes about 85% of Net Revenue. To compare the other countries more easily, it is left out of the top 10 countries by Net Revenue.

Year: All

Country: All

Top 10 countries by Net Revenue, outside the UK

United Kingdom excluded (about 85% of Net Revenue).



International orders are almost twice as large

Average order value. International = all countries except the United Kingdom.



Orders by weekday and hour

Darker = more orders. Columns are hours of the day (24h clock). Orders come mostly on weekdays around midday, and almost never on Saturdays.

Weekday	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mon		14	169	498	728	721	994	931	803	742	445	185	17		
Tue		17	204	622	806	825	1 107	1 038	839	811	469	294	81	31	2
Wed		11	202	552	836	829	1 212	1 041	941	773	511	203	1	2	1
Thu	1	22	210	545	848	813	1 140	987	898	840	649	517	360	288	49
Fri		12	207	497	819	801	937	799	813	591	315	180	6	9	1
Sat					6	1	6	7	3	5	2				
Sun				6	520	848	973	835	677	725	237	3			

Kamil Majeri / Source: Online Retail II (via Kaggle)

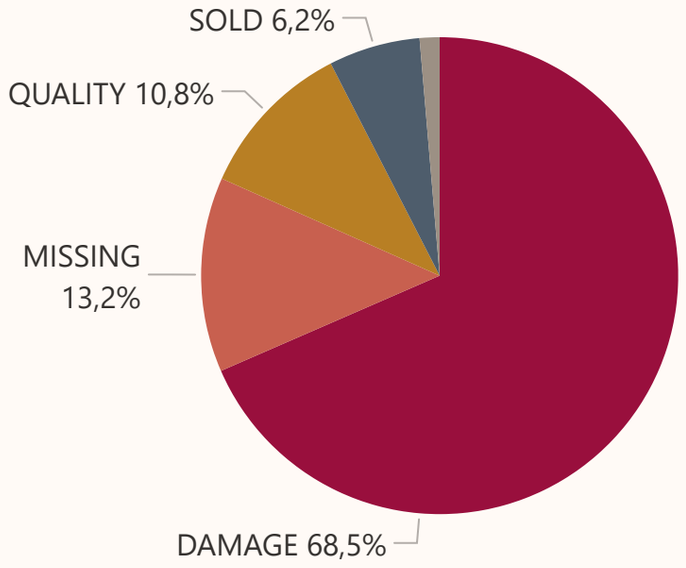
Can we trust the numbers?

Error lines have a price of 0, so they do not change revenue. Refunds are kept and shown separately from sales.

Year: All

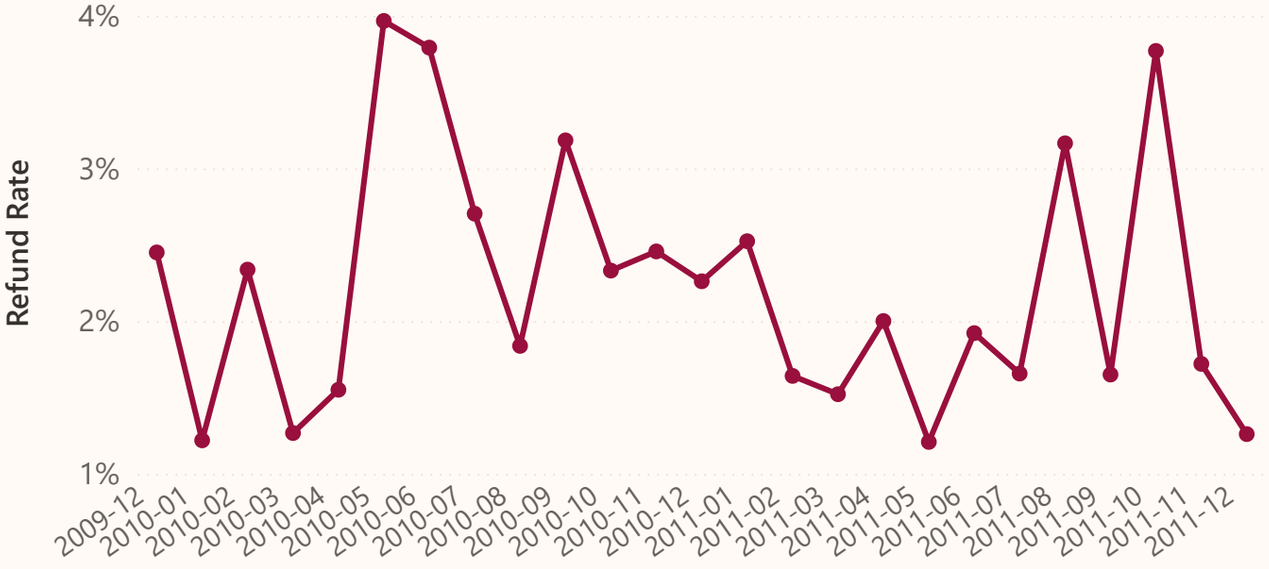
371	-£435 657	2,3%
Error Lines	Refunds	Refund Rate

Most error lines are damaged goods



Refunds stay below 4% of sales every month

Refunds as a share of gross revenue. Dec 2011 covers 1-9 Dec only.



Kamil Majeri / Source: Online Retail II (via Kaggle)